LAB 7: Fuzzing for Input Validation Bugs

Project: L6_L7_05 — Blood Donation Screening | **SRN:** PES2UG24CS191

Step 2: Running the Buggy Application (processor.py)

Command: python3 processor.py — Works on expected format but crashes with unhandled exceptions on unexpected inputs.

```
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 % python3 processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): O+
Validated: O Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/80
BP Metrics: Diff: 40, Ratio: 1.5

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D101|65.5|14.2
Status: Donor D101: Eligible

hemakshbreja@Hemakshs-MacBook-Pro Lab 7 % python3 processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): invalid
Traceback (most recent call last):
  File "/Users/hemakshbreja/SE/LAB6/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 50, in <module>
    main()
  File "/Users/hemakshbreja/SE/LAB6/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 39, in main
    print("Validated:", validate_blood_type(b_input))
  File "/Users/hemakshbreja/SE/LAB6/L6_L7_05_blood_donation_screening/Lab 7/processor.py", line 10, in validate_blood_type
    return f"{antigen} {rh_map[rh_char]}"
KeyError: 'D'
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 %
```

Step 4: Running Hypothesis Fuzz Tests on Buggy Application (Failing Test Cases)

Command: pytest test_processor.py — Hypothesis discovers multiple edge cases (None, empty strings, missing delimiters, invalid numbers) across all three functions.

```
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 % pytest test_processor.py
============================= test session starts ==============================
platform darwin -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/hemakshbreja/SE/LAB6/L6_L7_05_blood_donation_screening/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py FFF                                                    [100%]

=================================== FAILURES ===================================
_____________________ test_validate_blood_type_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | KeyError: '0'
    | AssertionError: validate_blood_type crashed with: '0'
    | Failing test case: test_validate_blood_type_no_crash(s='0')
    +---------------- 2 ----------------
    | IndexError: string index out of range
    | AssertionError: validate_blood_type crashed with: string index out of range
    | Failing test case: test_validate_blood_type_no_crash(s='')
    +---------------- 3 ----------------
    | AttributeError: 'NoneType' object has no attribute 'strip'
    | AssertionError: validate_blood_type crashed with: 'NoneType' object has no attribute 'strip'
    | Failing test case: test_validate_blood_type_no_crash(s=None)
_____________________ test_calculate_pulse_pressure_safe ______________________
  + Exception Group Traceback (most recent call last):
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_calculate_pulse_pressure_safe(s='/')
    +---------------- 2 ----------------
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_calculate_pulse_pressure_safe(s='')
    +---------------- 3 ----------------
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_calculate_pulse_pressure_safe(s=None)
_____________________ test_parse_donor_registry_safe ______________________
  + Exception Group Traceback (most recent call last):
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | ValueError: could not convert string to float: ''
    | Failing test case: test_parse_donor_registry_safe(s='|')
    +---------------- 2 ----------------
    | IndexError: list index out of range
    | Failing test case: test_parse_donor_registry_safe(s='')
    +---------------- 3 ----------------
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_donor_registry_safe(s=None)
=========================== short test summary info ============================
FAILED test_processor.py::test_validate_blood_type_no_crash - ExceptionGroup:...
FAILED test_processor.py::test_calculate_pulse_pressure_safe - ExceptionGroup...
FAILED test_processor.py::test_parse_donor_registry_safe - ExceptionGroup: Hy...
============================== 3 failed in 1.87s ===============================
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 %
```

Step 5: Running the Fixed Application (processor.py)

Command: python3 processor.py — Hardened functions handle both valid and malformed inputs gracefully without crashing.

```
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 % python3 processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): O+
Validated: O Positive

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/80
BP Metrics: Diff: 40, Ratio: 1.5

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): D101|65.5|14.2
Status: Donor D101: Eligible

hemakshbreja@Hemakshs-MacBook-Pro Lab 7 % python3 processor.py
==== Running validate_blood_type ====
Enter blood type (e.g. A+, O-, AB+): invalid
Validated: Invalid blood type

==== Running calculate_pulse_pressure ====
Enter BP (e.g. 120/80): 120/0
BP Metrics: Invalid BP values

==== Running parse_donor_registry ====
Enter record (ID|Weight|Hemoglobin): |
Status: Invalid donor record format
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 %
```

Step 6: Re-Running Hypothesis Fuzz Tests on Fixed Application (Passing)

Command: pytest test_processor.py — All property-based fuzz tests pass without any unhandled exceptions.

```
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 % pytest test_processor.py
============================= test session starts ==============================
platform darwin -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/hemakshbreja/SE/LAB6/L6_L7_05_blood_donation_screening/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 0.21s ===============================
hemakshbreja@Hemakshs-MacBook-Pro Lab 7 %
```